

Support Vector Machines (SVMs)

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Outline

1. Introduction
2. Separating hyperplane
3. Linear SVM with data can be separated
4. Linear SVM with data can not be separated – Soft margin SVM
5. Non-linear SVM
 - Kernel function
6. Conclusion

SVMs – Introduction (1)

- SVMs is proposed by V. Vapnik et al. in 1970s in Russia
- SVMs is a linear classifier for the purpose of defining a hyperplane to separate 2 classes of data (binary classifier). Then SVMs is improved for multiple-class classifier
- Kernel functions, also called transformation functions, are used for nonlinear classification cases
- SVM is a good (suitable) method for classification problems with large attribute representation spaces
- SVM is known as one of the best classification methods for text classification problems

SVMs – Introduction (2)

- Represent the set of r training examples:

$$\{ (\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_r, y_r) \}$$

- $\mathbf{x}_i \in \mathbf{R}^n$ is the input vector representing the training example
- y_i is a output label, $y_i \in \{1, -1\} = \{\text{positive class}, \text{negative class}\}$
- SVMs define a linear decomposition function:

$$f(\mathbf{x}) = \langle \mathbf{w} \cdot \mathbf{x} \rangle + b$$

- $\mathbf{w}$ weight vector corresponding to the attributes
- b is a real number (bias number)
- For each x_i :
$$y_i = \begin{cases} 1 & \text{if } \langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b \geq 0 \\ -1 & \text{if } \langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b < 0 \end{cases}$$

Outline

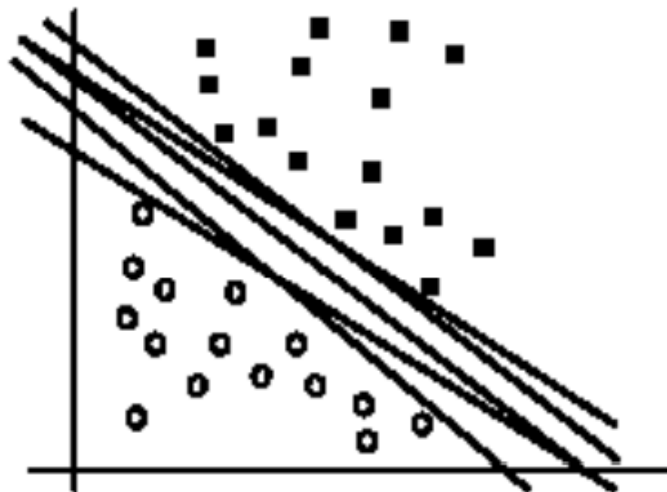
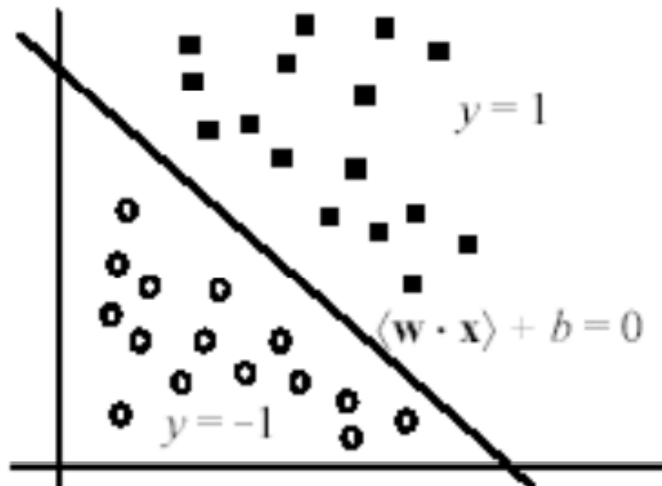
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Separating hyperplane

- The hyperplane separates the positive class training examples and the negative class training examples:

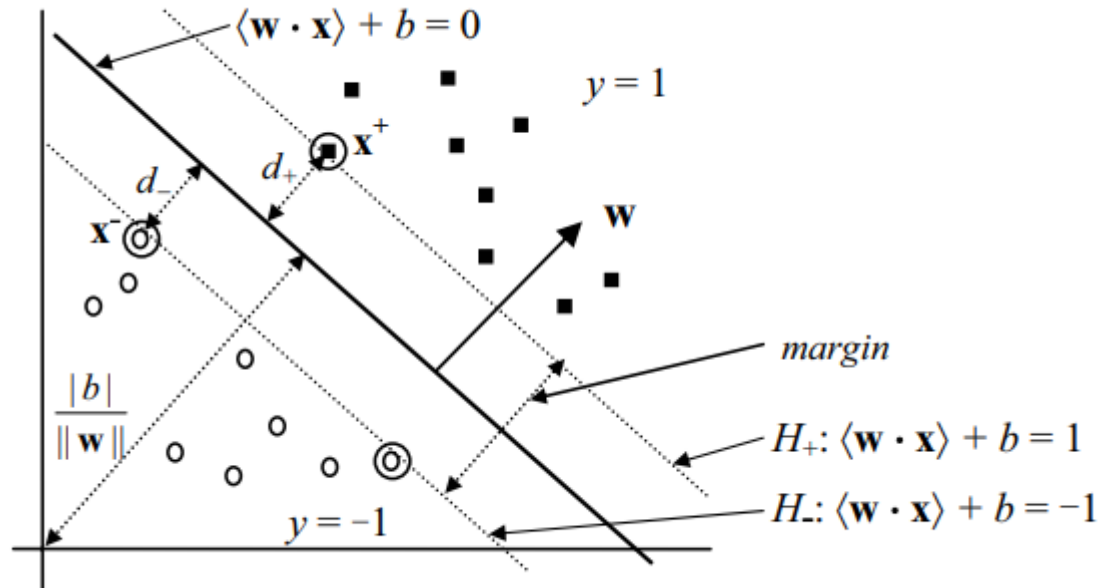
$$\langle \mathbf{w} \cdot \mathbf{x} \rangle + b = 0$$

- There are a lot of separating hyperplanes. Which one will be chosen



Hyperplane with maximum margin

- SVM selects the split hyperplane with the largest margin
- Machine learning theory shows that a largest margin separable hyperplane will minimize the error of classification



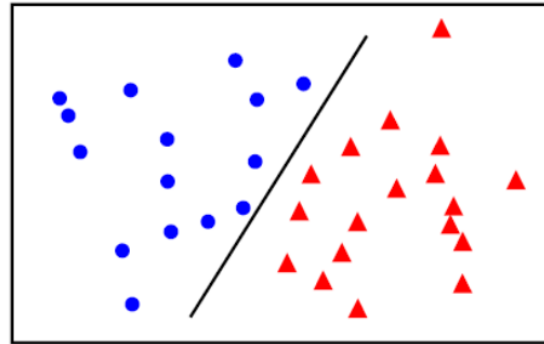
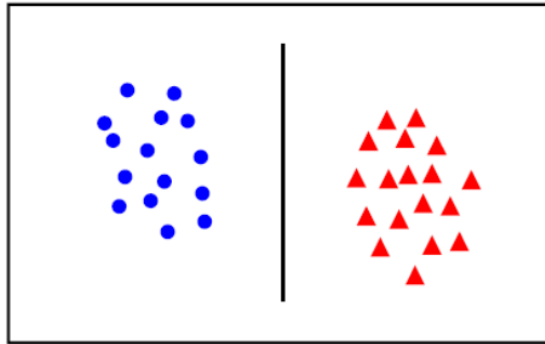
Outline

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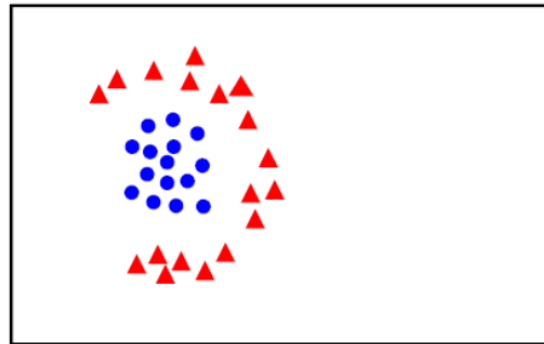
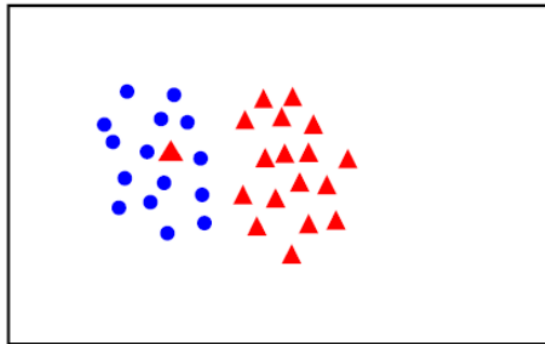
Data can be linearly separable (1)

- Assume that the data set can be linearly separable

linearly
separable



not
linearly
separable



Linear SVM with Data can be separated

- Consider a positive example $(\mathbf{x}^+, 1)$ and a negative example $(\mathbf{x}^-, 1)$ are nearest hyperlane H_0 ($\langle \mathbf{w} \cdot \mathbf{x} \rangle + b = 0$)
- Define 2 margin hyperlanes that parallel to each other
 - H_+ passes $\mathbf{x}^+$, and is parallel to H_0
 - H_- passes $\mathbf{x}^-$, and is parallel to H_0

$$H_+ : \langle \mathbf{w} \cdot \mathbf{x}^+ \rangle + b = 1$$

$$H_- : \langle \mathbf{w} \cdot \mathbf{x}^- \rangle + b = -1$$

so that:

$$\langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b \geq 1, \quad y_i = 1$$

$$\langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b \leq -1, \quad y_i = -1$$

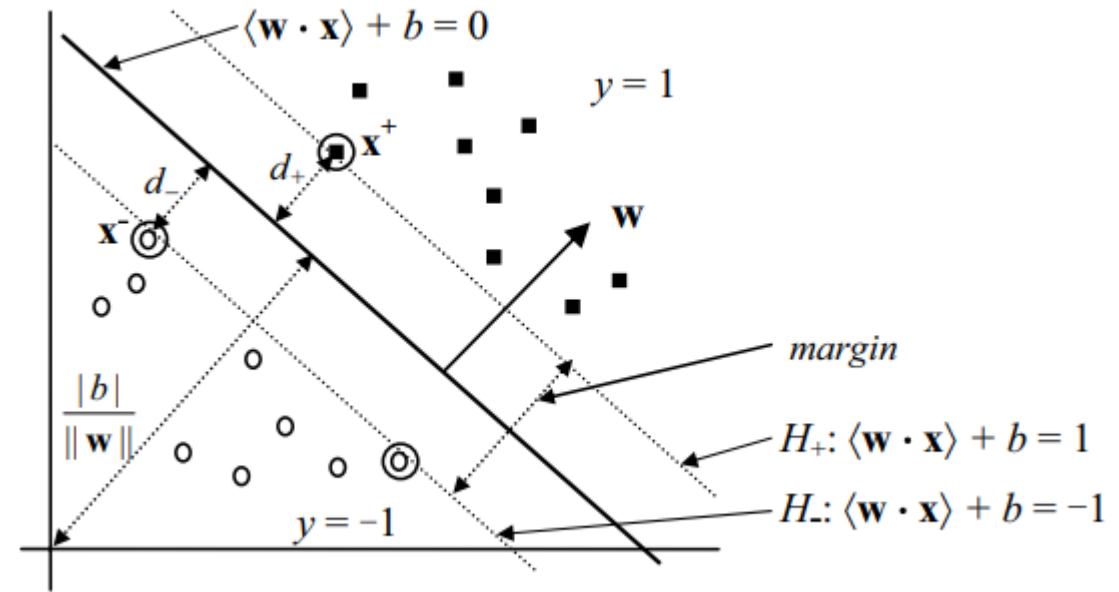
Calculate Margin Level (1)

- **Margin level** is the distance between two margin hyperplanes H_+ and H_- .

- d_+ is the distance between H_+ and H_0
- d_- is the distance between H_- and H_0
- $(d_+ + d_-)$ is margin level

- The distance from point x_i to the

plane H_0 is: $\frac{|\langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b|}{\|\mathbf{w}\|}$



where $\|\mathbf{w}\| = \sqrt{w_1^2 + w_2^2 + \dots + w_n^2}$ (the length of $\mathbf{w}$)

Calculate Margin Level (1)

- Find d_+ : distance from x^+ to $\langle \mathbf{w} \cdot \mathbf{x} \rangle + b = 0$

$$d_+ = \frac{|\langle \mathbf{w} \cdot \mathbf{x}_+ \rangle + b|}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}$$

- Find d_- : distance from x^- to $\langle \mathbf{w} \cdot \mathbf{x} \rangle + b = 0$

$$d_- = \frac{|\langle \mathbf{w} \cdot \mathbf{x}_- \rangle + b|}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}$$

- Find margin level:

- $margin = d_+ + d_- = \frac{2}{\|\mathbf{w}\|}$

SVM learning – Maximize Margin

Definition: (Linear SVM - The case of separable)

- The set of r linearly separable training examples:

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_r, y_r)\}$$

- SVM learns a classification to maximize margin
- Equivalent to solve the following optimization problem
 - Find $\mathbf{w}$ and b so that $margin = \frac{2}{\|\mathbf{w}\|}$ is maximum
 - with the conditions:

$$\begin{cases} \langle \mathbf{w}, \mathbf{x}_i \rangle + b \geq 1, & \text{if } y_i = 1 \\ \langle \mathbf{w}, \mathbf{x}_i \rangle + b \leq -1, & \text{if } y_i = -1 \end{cases}$$

Maximize Margin – Optimization problem

- SVM learning equivalent to :

- Minimize: $\frac{\langle \mathbf{w} \cdot \mathbf{w} \rangle}{2}$

- Condition: $\begin{cases} \langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b \geq 1, & \text{if } y_i = 1 \\ \langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b \leq -1, & \text{if } y_i = -1 \end{cases}$

- ... Equivalent to:

- Minimize: $\frac{\langle \mathbf{w} \cdot \mathbf{w} \rangle}{2}$

- Condition: $y_i(\langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b) \geq 1, \forall i = 1, \dots, r$

Constrained optimization theory (1)

- Minimization problem with equality constraint

Minimize $f(x)$ with condition $g(x)=0$

- Necessary condition for x_0 to be a solution:

$$\begin{cases} \frac{\partial}{\partial x} (f(x) + \alpha g(x)) \Big|_{x=x_0} = 0 \\ g(x) = 0 \end{cases} ; \text{ where } \alpha \text{ is a Lagrange coefficient}$$

- In the case of multi constraints:

$$\begin{cases} \frac{\partial}{\partial x} (f(x) + \sum_{i=1}^r \alpha_i g_i(x)) \Big|_{x=x_0} = 0 \\ g_i(x) = 0 \end{cases} ; \text{ where } \alpha_i \text{ is Lagrange coefficients}$$

Constrained optimization theory

- Minimization problem with inequality constraint

Minimize $f(x)$ with condition $g_i(x) \leq 0$

- Necessary condition for x_0 to be a solution:

$$\begin{cases} \left. \frac{\partial}{\partial x} (f(x) + \sum_{i=1}^r \alpha_i g_i(x)) \right|_{x=x_0} = 0; & \text{where } \alpha \geq 0 \text{ is a Lagrange coefficient} \\ g_i(x) \leq 0 \end{cases}$$

- Hàm Lagrange:

$$L = f(x) + \sum_{i=1}^r \alpha_i g_i(x)$$

$$\Rightarrow L_p(w, b, \alpha) = \frac{\langle w \cdot w \rangle}{2} + \sum_{i=1}^r \alpha_i [y_i (\langle w \cdot x_i \rangle + b) - 1]$$

where $\alpha_i \geq 0$ are Lagrange coefficients

Solve the constrained minimization problem (1)

- **Optimization theory** : an optimal solution for above equation must satisfy certain conditions, called the Karush-Kuhn-Tucker (KKT) conditions – which are necessary conditions (but not the sufficient condition)

$$(1) \quad \frac{\partial L_P}{\partial \mathbf{w}} = \mathbf{w} - \sum_{i=1}^r \alpha_i y_i \mathbf{x}_i = 0$$

$$(2) \quad \frac{\partial L_P}{\partial b} = -\sum_{i=1}^r \alpha_i y_i = 0$$

$$(3) \quad y_i (\langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b) - 1 \geq 0, \quad \forall \mathbf{x}_i \quad (i = 1..r)$$

$$(4) \quad \alpha_i \geq 0$$

$$(5) \quad \alpha_i (y_i (\langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b) - 1) = 0$$

- (5) show that only $\mathbf{x}_i$ belong to margin hyperlanes (H_+ , H_-) have $\alpha_i > 0$ (because $y_i (\langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b) = 1$ in this case) => **those $\mathbf{x}_i$ are called support vectors**
 - $\alpha_i = 0$ with other $\mathbf{x}_i$

Solve the constrained minimization problem (2)

- For this minimization problem under consideration with a convex objective function and linear constraints, the conditions KKT are necessary and sufficient for an optimal solution.
- It is difficult to deal with inequality constraint, so that we use Lagrange method to leads to a dual expression of the optimization problem:

$$L_D(\boldsymbol{\alpha}) = \sum_{i=1}^r \alpha_i - \frac{1}{2} \sum_{i,j=1}^r \alpha_i \alpha_j y_i y_j \langle \mathbf{x}_i \cdot \mathbf{x}_j \rangle$$

=> The solution is found by minimize L_p or maximize L_D

The dual optimization problem

Maximize:
$$L_D(\boldsymbol{\alpha}) = \sum_{i=1}^r \alpha_i - \frac{1}{2} \sum_{i,j=1}^r \alpha_i \alpha_j y_i y_j \langle \mathbf{x}_i \cdot \mathbf{x}_j \rangle$$

Conditions:
$$\begin{cases} \sum_{i=1}^r \alpha_i y_i = 0 \\ \alpha_i \geq 0, \forall i = 1..r \end{cases}$$

- For a convex objective function and linear constraints, the maximum value of L_D occurs at the same values of $\mathbf{w}$, $\mathbf{b}$ and $\boldsymbol{\alpha}_i$ which helps to obtain minimum value of L_P
- Solve the above equation will get $\boldsymbol{\alpha}_i$ (out of our lecture), then help to find $\mathbf{w}^*$, $\mathbf{b}^*$

Find $\mathbf{w}^*$ and b^*

- Let **SV** be the set of support vectors
 - SV is a subset of the set of r initial training examples
 - > $\alpha_i > 0$ for support vectors $\mathbf{x}_i$
 - > $\alpha_i = 0$ for vectors are not support vectors $\mathbf{x}_i$

- Use KKT (1) we have

$$\mathbf{w}^* = \sum_{i=1}^r \alpha_i y_i \mathbf{x}_i = \sum_{\mathbf{x}_i \in \text{SV}} \alpha_i y_i \mathbf{x}_i$$

- Consider KKT (5) and any support vector $\mathbf{x}_k$, we have:

$$y_k (< \mathbf{w}^* . \mathbf{x}_k > + b^*) - 1 = 0$$

$$\Leftrightarrow y_k (< \mathbf{w}^* . \mathbf{x}_k > + b^*) = 1$$

$$\Leftrightarrow b^* = y_k - (< \mathbf{w}^* . \mathbf{x}_k >) \quad (\text{because } y_k^2 = 1)$$

Determining the classification

- Classification is determined by the hyperplane:

$$f(\mathbf{x}) = \langle \mathbf{w}^*, \mathbf{x} \rangle + b^* = \sum_{\mathbf{x}_i \in SV} \alpha_i y_i \langle \mathbf{x}_i, \mathbf{x} \rangle + b^* = 0 \quad (\text{I})$$

- For a new example $\mathbf{z}$, we need to define:

$$\text{sign}(\langle \mathbf{w}^*, \mathbf{z} \rangle + b^*) = \text{sign}(\sum_{\mathbf{x}_i \in SV} \alpha_i y_i \langle \mathbf{x}_i, \mathbf{z} \rangle + b^*) \quad (\text{II})$$

$\Rightarrow$ If (II) return 1 then example $\mathbf{z}$ is classified into the positive class; otherwise, it is classified into the negative class

- Comment:
 - This classification only depends on support vectors
 - Only need the value of the dot product of 2 vectors (not needing to know the value of those 2 vectors)

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Linear SVM with Data can not be separated

- The case of linear and separable classifiers is ideal (at least happen)
- The data set may contain noise and errors (e.g. some examples are mislabeled)
- If the data set contains noise, the conditions may not be satisfied
 - Can not find w^* and b^* satisfied $y_i(< \mathbf{w} \cdot \mathbf{x}_i > + b) \geq 1$
- To work with noisy data, it is necessary to loosen margin constraints by using slack variables ϵ_i

⇒ The new condition:

$$y_i(< \mathbf{w} \cdot \mathbf{x}_i > + b) \geq 1 - \xi_i \quad \forall i = 1, \dots, r$$

$$\xi_i \geq 0$$

⇒ We have new optimization problem – soft margin SVM

Soft Margin SVM

Minimize: $\frac{\langle \mathbf{w} \cdot \mathbf{w} \rangle}{2} + C \sum_{i=1}^r \xi_i$ (C is called error cost)

Condition:
$$\begin{cases} y_i(\langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b) \geq 1 - \xi_i & \forall i = 1, \dots, r \\ \xi_i \geq 0 \end{cases}$$

- Lagrange optimal expression:

$$L_p = \frac{\langle \mathbf{w} \cdot \mathbf{w} \rangle}{2} + C \sum_{i=1}^r \xi_i - \sum_{i=1}^r \alpha_i [y_i(\langle \mathbf{w} \cdot \mathbf{x}_i \rangle + b) - 1 + \xi_i] - \sum_{i=1}^r \mu_i \xi_i$$

where $\alpha_i (\geq 0)$ and $\mu_i (\geq 0)$ are multi coefficient Lagrange

The dual optimization problem

Maximize:
$$L_D(\boldsymbol{\alpha}) = \sum_{i=1}^r \alpha_i - \frac{1}{2} \sum_{i,j=1}^r \alpha_i \alpha_j y_i y_j \langle \mathbf{x}_i \cdot \mathbf{x}_j \rangle$$

Conditions:
$$\begin{cases} \sum_{i=1}^r \alpha_i y_i = 0 \\ 0 \leq \alpha_i \leq C, \quad \forall i = 1..r \end{cases}$$

=> Objective function is the same with separable linear classification

=> The only difference is: $\alpha_i \leq C$

Important characteristics from sprasity

- The solution is determined based on very few (sparse) α_i values
 - A lot of the learning examples are outside the margin area and they have $\alpha_i = 0$
 - The examples locate on the margin $y_i(< \mathbf{w} \cdot \mathbf{x}_i > + b) = 1$ (are the support vectors), then have $\alpha_i \neq 0$ ($0 < \alpha_i < C$)
 - Examples that fall within the margin $y_i(< \mathbf{w} \cdot \mathbf{x}_i > + b) < 1$ (are noisy/error examples), have $\alpha_i \neq 0$ ($\alpha_i = C$)
- Without this sparsity feature, the SVM method cannot be effective for large data sets.

The Classifier hyperplane

- Classification is determined by the hyperplane:

$$f(\mathbf{x}) = \langle \mathbf{w}^*, \mathbf{x} \rangle + b^* = \sum_{x_i \in SV} \alpha_i y_i \langle \mathbf{x}_i, \mathbf{x} \rangle + b^* = 0$$

$\Rightarrow$ A lot of $\mathbf{x}_i$ have $\alpha_i = 0$ (that is sparsity characteristic of SVMs)

- We determine the class for a new example $\mathbf{z}$ by:

$$\text{sign}(\langle \mathbf{w}^*, \mathbf{z} \rangle + b^*)$$

$\Rightarrow$ To find b^* need to identify C (in objective function)

- Often use validation set

Linear SVM - Conclusions

- Classification is based on the separating hyperplane
- The split hyperplane is defined based on the set of support vectors
- Only for support vectors, their Lagrange multiplier is $\neq 0$
 - For other training examples (which are not support vectors) their Lagrange multiplier = 0
- Determining the support vectors (among the training examples) requires solving a quadratic optimization problem
- In the dual expression (L_D) and in the separating hyperplane expression, the training examples appear only inside the inner/dot-products of the vectors.

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6. Conclusion

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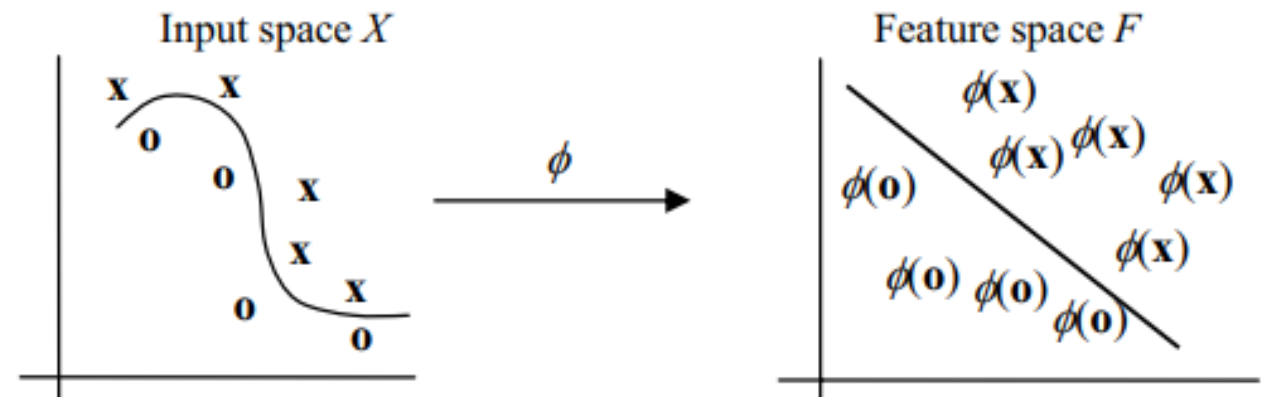
- The formulas in the SVM method require the data set to be linearly classifiable (with/without noise)
- In many practical problems, data sets can not be linearly separable
- Non-linear SVM method:
 - Phase 1: Convert the original input representation space to another space (usually with much higher dimensions)
 - > The data is represented in the new space can be linearly separable
 - Phase 2: Re-apply the formulas and steps as in the linear SVM classification method

Transforming performance space

- The idea is to map (transform) the data representation from the original space X to another space F by applying a nonlinear mapping function ϕ
- In the transformed space, the set of original learning examples $\{(x_1, y_1), (x_2, y_2), \dots, (x_r, y_r)\}$ will be performed:
 $\{(\phi(x_1), y_1), (\phi(x_2), y_2), \dots, (\phi(x_r), y_r))\}$
- Example: $(x_1, x_2) \rightarrow (x_1^2, x_2^2, x_1 x_2)$
 $(x = (2, 3), y = -1) \Rightarrow (x = (4, 9, 6), y = -1)$

$\Rightarrow$ Problem: dimension is too big

$\Rightarrow$ Use Kernel function



Kernel functions

- Consider the classification hyperplane:

$$f(\mathbf{z}) = \langle \mathbf{w}^*, \phi(\mathbf{z}) \rangle + b^* = \sum_{i=1}^r \alpha_i y_i \langle \phi(\mathbf{x}_i), \phi(\mathbf{z}) \rangle + b^*$$

$\Rightarrow$ Not need to find exactly $\phi(\mathbf{x}_i)$ and $\phi(\mathbf{z})$

$\Rightarrow$ Only need to find inner product $\langle \phi(\mathbf{x}_i), \phi(\mathbf{z}) \rangle$

$\Rightarrow$ Use Kernel functions K :

$$K(\mathbf{x}, \mathbf{z}) = \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle$$

$\Rightarrow$ We do not need to know what ϕ is

Kernel functions - Example

- Polynomial Kernel function:

$$K(\mathbf{x}, \mathbf{z}) = \langle \mathbf{x} \cdot \mathbf{z} \rangle^d$$

- Consider $d=2$, $\mathbf{x} = (x_1, x_2)$ and $\mathbf{z} = (z_1, z_2)$:

$$\begin{aligned}\langle \mathbf{x} \cdot \mathbf{z} \rangle^2 &= (x_1 z_1 + x_2 z_2)^2 \\ &= x_1^2 z_1^2 + 2x_1 z_1 x_2 z_2 + x_2^2 z_2^2 \\ &= \langle (x_1^2, x_2^2, \sqrt{2}x_1 x_2) \cdot (z_1^2, z_2^2, \sqrt{2}z_1 z_2) \rangle \\ &= \langle \phi(\mathbf{x}) \cdot \phi(\mathbf{z}) \rangle = K(\mathbf{x}, \mathbf{z})\end{aligned}$$

- The above example shows that the kernel function $\langle \mathbf{x} \cdot \mathbf{z} \rangle^2$ is a dot product of two vectors $\phi(\mathbf{x})$ and $\phi(\mathbf{z})$ in the space after transformation

Commonly used kernel functions

- Polynomial:

$$K(\mathbf{x}, \mathbf{z}) = (\langle \mathbf{x}, \mathbf{z} \rangle + \theta)^d, \quad \theta \in R, d \in N$$

- Gaussian radial basis function (RBF)

$$K(\mathbf{x}, \mathbf{z}) = e^{-\frac{\|\mathbf{x} - \mathbf{z}\|^2}{2\sigma}}, \quad \sigma > 0$$

- Sigmoidal

$$K(\mathbf{x}, \mathbf{z}) = \tanh(\beta \langle \mathbf{x}, \mathbf{z} \rangle - \gamma) = \frac{1}{1 + e^{-(\beta \langle \mathbf{x}, \mathbf{z} \rangle - \gamma)}} \quad ; \beta, \gamma \in R$$

SVMs - Conclusions

- SVM only works with an input space of real numbers
 - For nominal attributes, it is necessary to convert the nominal values into numeric values
- SVM only works with 2-class classification
 - For multi-class classification problems, it is necessary to convert to a set of 2-class classification problems, and then solve each of these 2-class problems separately.
 - Eg: One vs All, One vs One
- Separated Hyperplane is difficult to understand especially Kernel function is used:
 - SVM is often used in application problems where explaining the behavior (decisions) of the system to the user is not an important requirement.

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Q&A